Logic used to construct baseline threads
Supporting Document`IndigoCarbon_US-1_2023_0048`

June 28, 2024

Contents

1 Key Definitions **1**

2 Summary **2**

3 How management event data meet SEP requirements for baselines **3**

4 How baseline construction permanently locks to field area using management event boundaries (Steps 1 – 3, Fig 1) **3**

5 Meeting field-level SEP requirements for baseline construction (Step 4, Fig 1) **7**

6 Constructing modeled baselines for individual sample points (Step 5, Fig 1) **9**

7 Reasoning for only using blended baselines **11**

8 Summary of Changes **14**

1 Key Definitions

1. **Field Boundary:** geospatial boundary of an individual field per SEP Section 2.2.2 defining the field-level project area being quantified for this reporting period as described in `IndigoCarbon_US-1_2023_0034`. See Figure 1 & 2
2. **Management Event:** a grouping of required management data for modeling and credit quantification, designed to facilitate grower data entry and assessment of SEP data requirements per SEP Section 6.1 as described in IndigoCarbon_US-1_2023_0032. Each *Management Event* has a defined *Management Event Boundary* that can be all or part of the field area defined by *Field Boundary*. There are no limits on the number of *Management Events* that can be defined for an individual field. See Figure 2.
3. **Management Event Boundary:** geospatial boundary of an individual *Management Event* that occurred within an individual *Field Boundary* during either the required historical baseline period or the with-project period. *Management Event Boundaries* are always associated with *Field Boundaries*. Each individual *Management Event Boundary* is permanently locked to that geospatial field area. *Management Event Boundaries* may be the same size as field boundaries, or they may be smaller. See Figure 2.
4. **Management Zones:** An area of an individual *Field Boundary* defined by the combination of *Management Event Boundaries* that describe the presence or absence of *Management Events* at any particular time or time window, as well as attributes of the *Management Events* (if any occurred). Concretely, *Management Zones* are a data-rich geospatial object created for each field using an algorithm that crawls through all *Management Events* and decides how to group them into discrete zonal areas based on their unique combinations. The creation of *Management Zones* enabling prorating of credit quantification for the area within the *Field Boundary* based on the geospatial boundaries of *Management Events*. See Figures 4, 5 and 6.

5. **Zone-Cycle:** A single cultivation cycle on a single *Management Zone* within a single field, considered collectively as a pair that define the 'atomic unit' of quantification for a given field in a given reporting period. See Figures 5 and 6.

2 Summary

Figure 1 provides a high-level summary of steps for how modeled baselines are constructed for the field area of an individual field, using the *field boundary* (Step 1), spatially and temporally explicit *management event boundaries* (Steps 2 – 3), and *initial soil sample points* (Step 4).

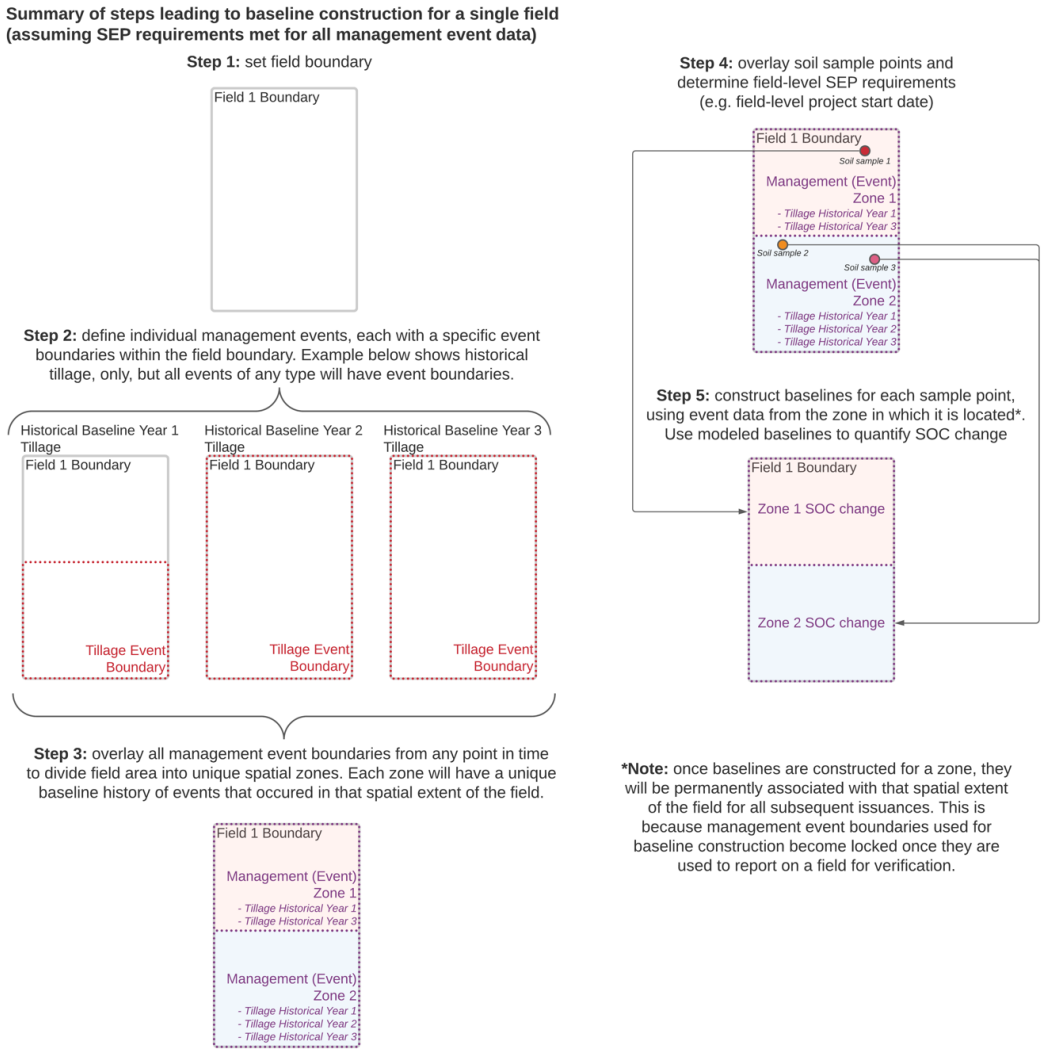


Figure 1: Simplified visualization of steps leading to the construction of baselines for modeling and quantifying SOC emissions reductions within zones of unique management practices for an individual field, using spatially explicit management event data.

Management Zones, created from *management event boundaries* as shown in Figure 1, Step 3 (described in detail in Section 2) are a critical component for many steps of the Carbon Data Pipeline, including to prorate the quantification of total field area based on the areal extent of unique combinations of management events. Once a Management Zone is used to construct a set of baselines, these baselines are locked permanently to the field area defined by the Management Zone.

In this document, Section 1 provides details for how management events meet SEP baseline data requirements. Section

2 describes how baseline construction accounts for field area using management event boundaries (Steps 1 – 3 above). Section 3 describes how field-level SEP requirements for constructing baseline threads are met (Step 4 above). Section 4 then provides the logic for baseline construction for an individual sample point (Step 5 above). Section 5 provides supporting material for baseline construction logic.

3 How management event data meet SEP requirements for baselines

The data used to construct baselines meet SEP data hierarchy requirements (SEP Section 6.1), by passing Grower Data QAQC (Gate 2 of the Carbon Data Pipeline, described in IndigoCarbon_US-1_2023_0032) and gap-filling (Gate 5 of the Carbon Data Pipeline, described in IndigoCarbon_US-1_2023_0031). As detailed in IndigoCarbon_US-1_2023_0032, Gate 2 is designed with more rigorous requirements for data associated with management events that are designated as Practice Changing- i.e. quantifiably associated with adopted practices that qualify a field for credit quantification. This helps ensure rigor and quality in the data used to construct baselines.

The data used to construct baselines meets SEP requirements for the length of the required historical baseline period (SEP Section 3.4.1.3) by passing Gate 2, Level 1 event establishment, as described in IndigoCarbon_US-1_2023_0032.

Management events are a software design construct used to group data and evidence and associate them with their own geospatial boundaries within an individual field boundary (see Figure 2 for examples). This makes the management data and evidence permanently associated with the area of the field contained within the management event boundary once they pass all SEP requirements (per SEP Section 6.1). This feature enables baselines construction to be permanently associated with the spatial extent of unique sets of management events within an individual field, as the management event boundaries will never change. Management event boundaries may be the same as the field boundary, or they may be smaller (Figure 2).

4 How baseline construction permanently locks to field area using management event boundaries (Steps 1 – 3, Fig 1)

At several stages in the Carbon Data Pipeline the boundaries of management events are used to create Management Zones within a field. Management Zones are data-rich, spatially-explicit objects created for each field using all management event data available at the point-in-time that Management Zones are created. They are generated for each individual field using an algorithm that crawls through all management events and decides how to group them into discrete zonal areas based on their unique combinations. The creation of Management Zones enables prorating of credit quantification for the area within the field boundary based on the geospatial boundaries of management events. Figure 3 shows three examples of Management Zone creation, demonstrating how changes in management event boundaries, alone, can change the number of Management Zones that are generated for an individual field area.

SEP requirements are assessed for each unique area of the field within each Management Zone- i.e. at the Management Zone level. For example, additionality is assessed for each Management Zone, based on the unique management events that occurred in that Management Zone. Management Zones are then used to assess field-level SEP requirements. For example, if some Management Zones qualify for additionality but others do not within an individual field, the area of the qualifying Management Zones (as compared to the total field area) will determine whether the field meets minimum field-level requirements for inclusion in the project (see IndigoCarbon_US-1_2023_0032 for details).

After gap-filling, grower data QAQC, and grower attestation are complete, management events boundaries will receive no further additions or modifications (unless an error is found), allowing baseline construction to lock permanently to the spatial extents of historical baseline management events. Constructing baselines using Management Zones allows our quantification methods to prorate according to the explicit spatial area of unique management event combinations. Note that once event boundaries become locked for a given reporting period, the Management Zones for a field can only become further subdivided for future reporting periods, as the event boundaries in previous reporting periods will never change (Figure 4).

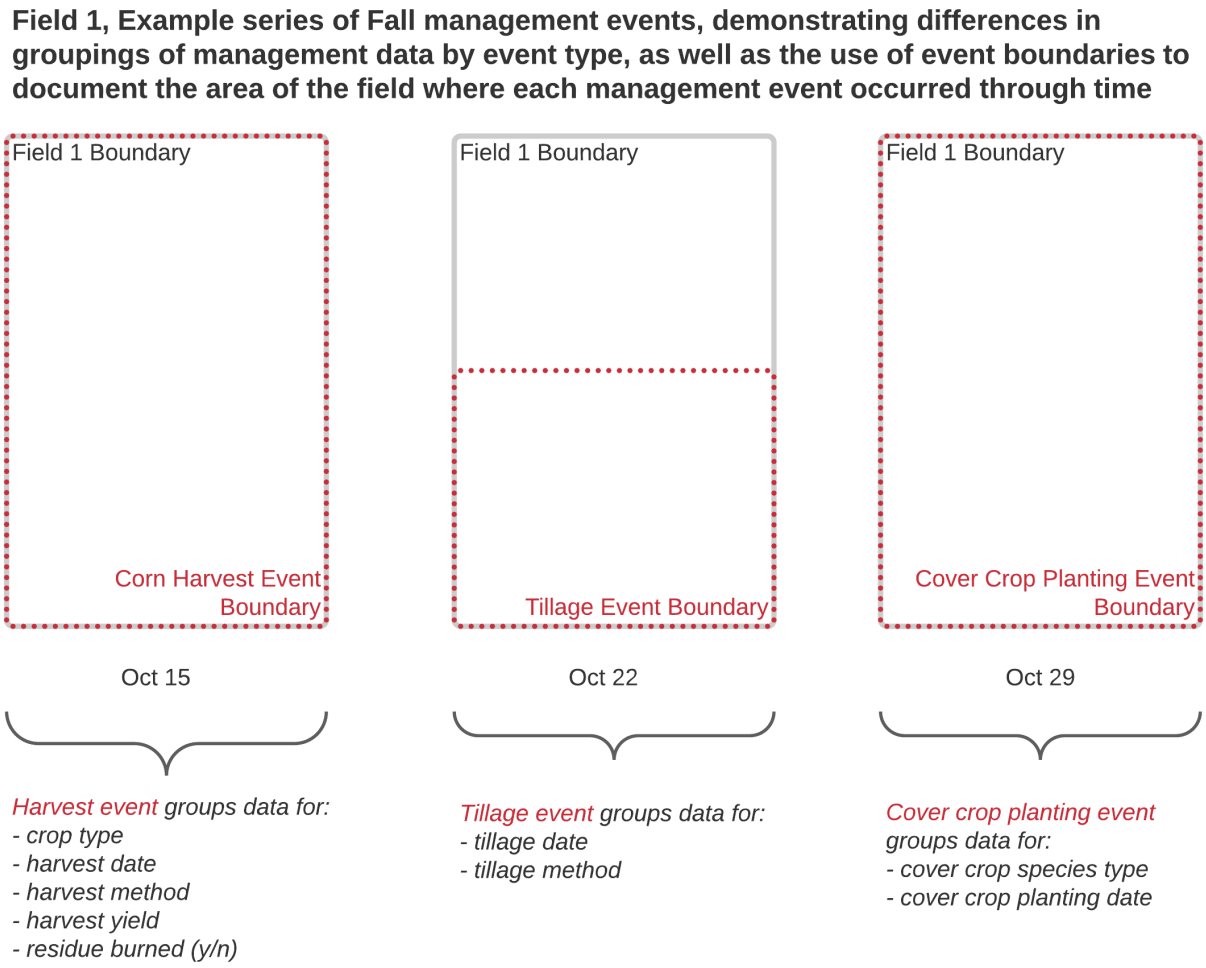


Figure 2: Example of management event boundaries (red), as compared to the field boundary (grey), for a series of fall management practices in a single field. Once attested, the management data and evidence collected for each event (bottom row) are permanently associated with the specific area of the field defined by the boundary of that event. See IndigoCarbon_US-1_2023_0032 for details.

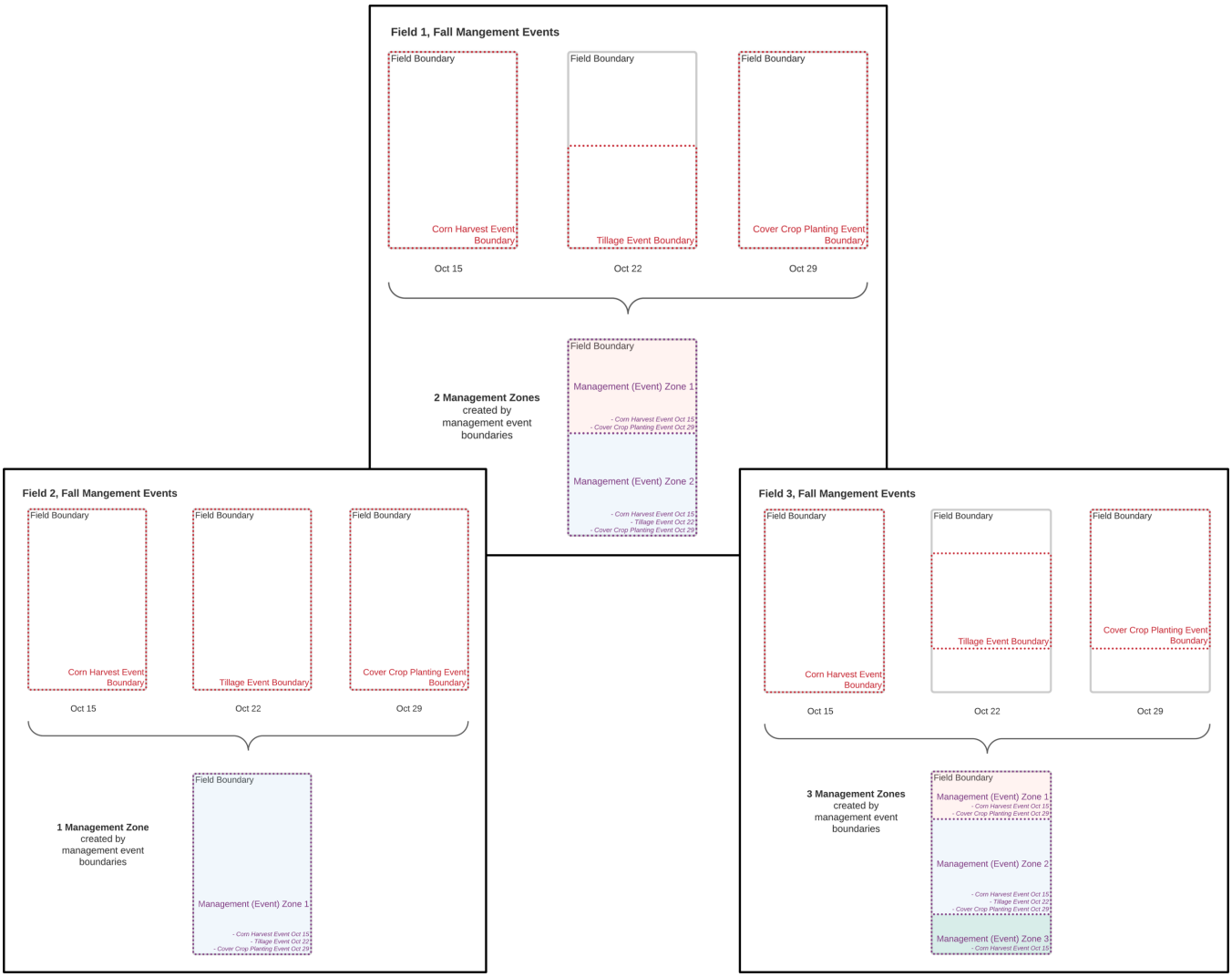


Figure 3: Simplified example of how the algorithm for creating Management Zones would generate two (Field 1), one (Field 2) or three (Field 3) Management Zones for individual fields, even when the management data between fields is the same and only the management event boundaries vary.

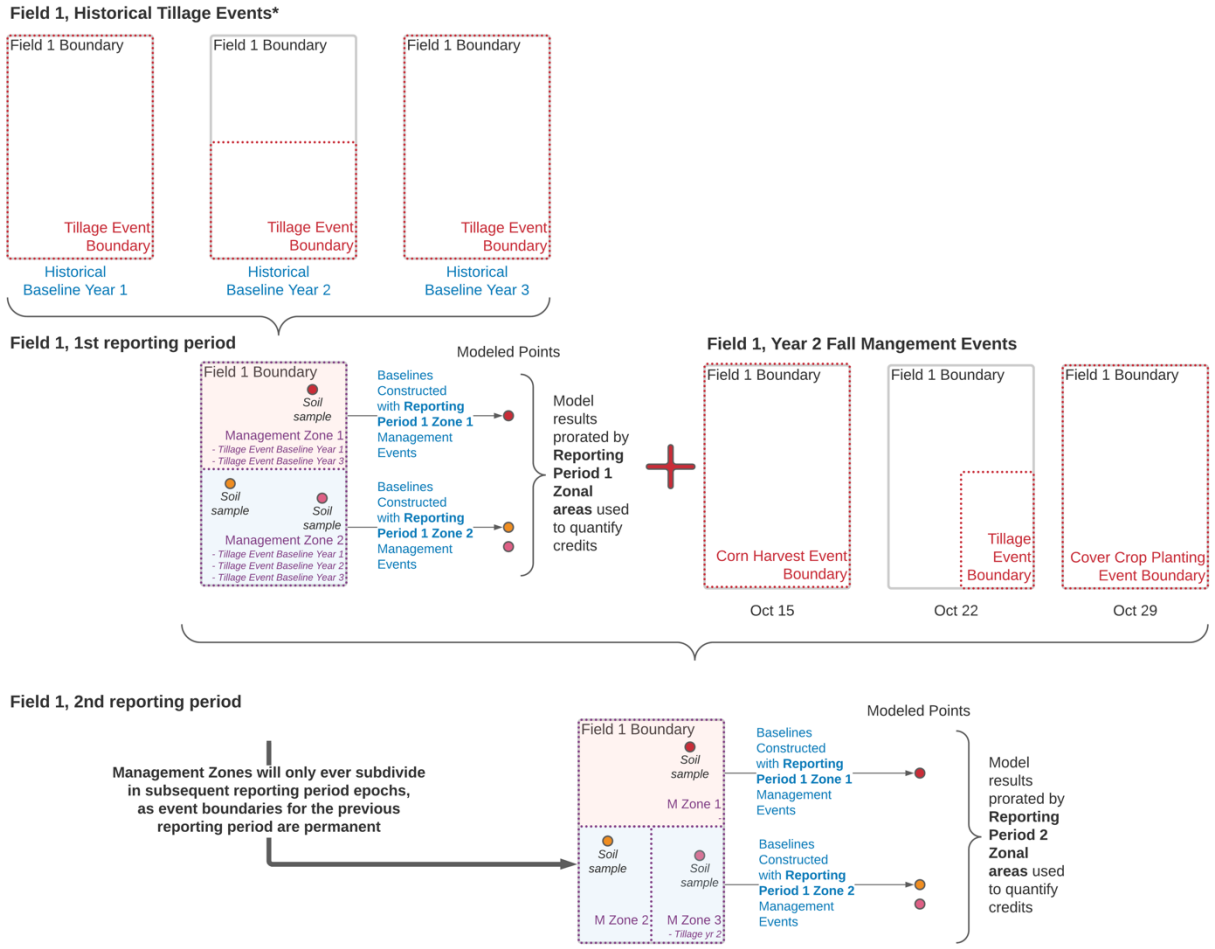


Figure 4: Simplified example of how event boundaries lock baseline construction (blue text) permanently to unique zonal areas within an individual field (marked by pink vs blue areas in the 2nd and 3 rows), even if with-project event boundaries vary in the future (bottom row, noting the unique tillage event boundary that resulted in subdividing the field area). This approach allows the same baselines to always be used for any quantification of the zonal area for which they have been constructed.

5 Meeting field-level SEP requirements for baseline construction (Step 4, Fig 1)

Per SEP requirements in Section 7.2, and as described in IndigoCarbon_US-1_2023_0016 *Cultivation cycles and crop growing seasons,* agricultural events – harvest events, when possible- are used to discretize time.

Therefore, in the Carbon Data Pipeline, the spatial boundaries of harvest events - in some circumstances also planting events and fallow periods (i.e. the lack of events) - play an important role in dividing not only space, but also time. To be consistent with SEP requirements described in SEP Section 7.2, cultivation cycles are determined for each individual Management Zone. This ensures that the division of time stays aligned with the prorating of area in the quantification of events, and further keeps time divisions both agronomically sound and spatially consistent.

A single cultivation cycle on a Management Zone, considered collectively as a pair (a "zone–cycle"), is the 'atomic unit' of quantification for a given field in a given reporting period. Per SEP requirements in Section 5.1, this supports modeling a baseline for each 'zone-cycle' of the crediting period with complete and qualifying management event data.

SEP requirements for the project start date, per SEP Section 3.2, are met across atomic units as follows:

- The ***project-level start date***, is defined as the earliest qualifying ***field-level start date*** in the project. Therefore, reported credits for any field will never begin before the ***project-level start date***.
- The ***field-level start date*** is defined as the day starting the earliest zone-cycle (across all Management Zones within an individual field) that meets SEP requirements for additionality. Therefore, an individual field will never report credits before the ***field-level start date***.
- The ***Management Zone-level start date*** is not specified in the SEP, but conceptually applies SEP Section 3.2 requirements and is defined as the day starting the first zone-cycle that meets SEP requirements for additionality (reminder: additionality is assessed at the Management Zone level, see IndigoCarbon_US-1_2023_0032 for details).

Any difference between the ***field-level start date*** and ***Management Zone-level start date*** is handled by delaying the start of reporting for that portion of the field (see Figure 5 for an example). This ensures any discrepancies between the two only ever lead to fewer credits by reducing the time that credits are quantified on that zonal area.

Per SEP 7.2, only completed cultivation cycles are used to quantify credits for project reporting. In our project, zone-cycle atomic units are used to account for the zonal area of completed cultivation cycles quantified in a given reporting period. To ensure that the same zone-cycle start and end dates are used for all aspects of quantification, a Management Zone Attribute Table (ZAT) is used as a common reference. The ZAT is a relational table with a row per cultivation cycle/baseline thread (baseline threads as described in Section 4 below) per management zone. Each row has the necessary data to identify a management zone and the field it is associated with along with the start and end dates of the cycle/thread. Each row is amended with additional information about the specific zone and time period to be used in subsequent quantification steps such as soil composition and climatology data. The ZAT is used to account for field area and the start and end dates of the zone-cycles reported in the data submission file.

Minor human errors in drawing sub-field event boundaries (i.e., boundaries that do not perfectly cover 100% of the field area) were permitted in some circumstances, as long as event boundaries covered at least 95% of the total field (see IndigoCarbon_US-1_2023_0032 for details). Due to software limitations, this rule allowed a small portion of the total project area to have insufficient data for quantification in this reporting period- i.e., these areas are not included in the ZAT.

The Management Zone algorithm can also create very small slivers of field areas[1] that are excluded from or duplicated in Management Zones, mainly due to imprecisions from imported data sources and hand-drawn boundaries (especially prior to the addition of the auto-snapping feature described above), but also due to accumulation of numerical errors from repeated intersection and geospatial operations.

The net impact of the Management Zone algorithm is immaterial; less than 0.1% of total project acres are affected by the combined impacts of the 95% rule for event boundaries and Management Zone computation errors. Steps have been taken to minimize Management Zone computational errors, including de-duplication of zonal areas across time prior to credit quantification. In subsequent reporting periods, the mitigation steps described above will help address

[1] Slivers were on the order of ~0.05 acres on average per affected field when assessed Nov. 10, 2021, prior to several mitigation corrections.

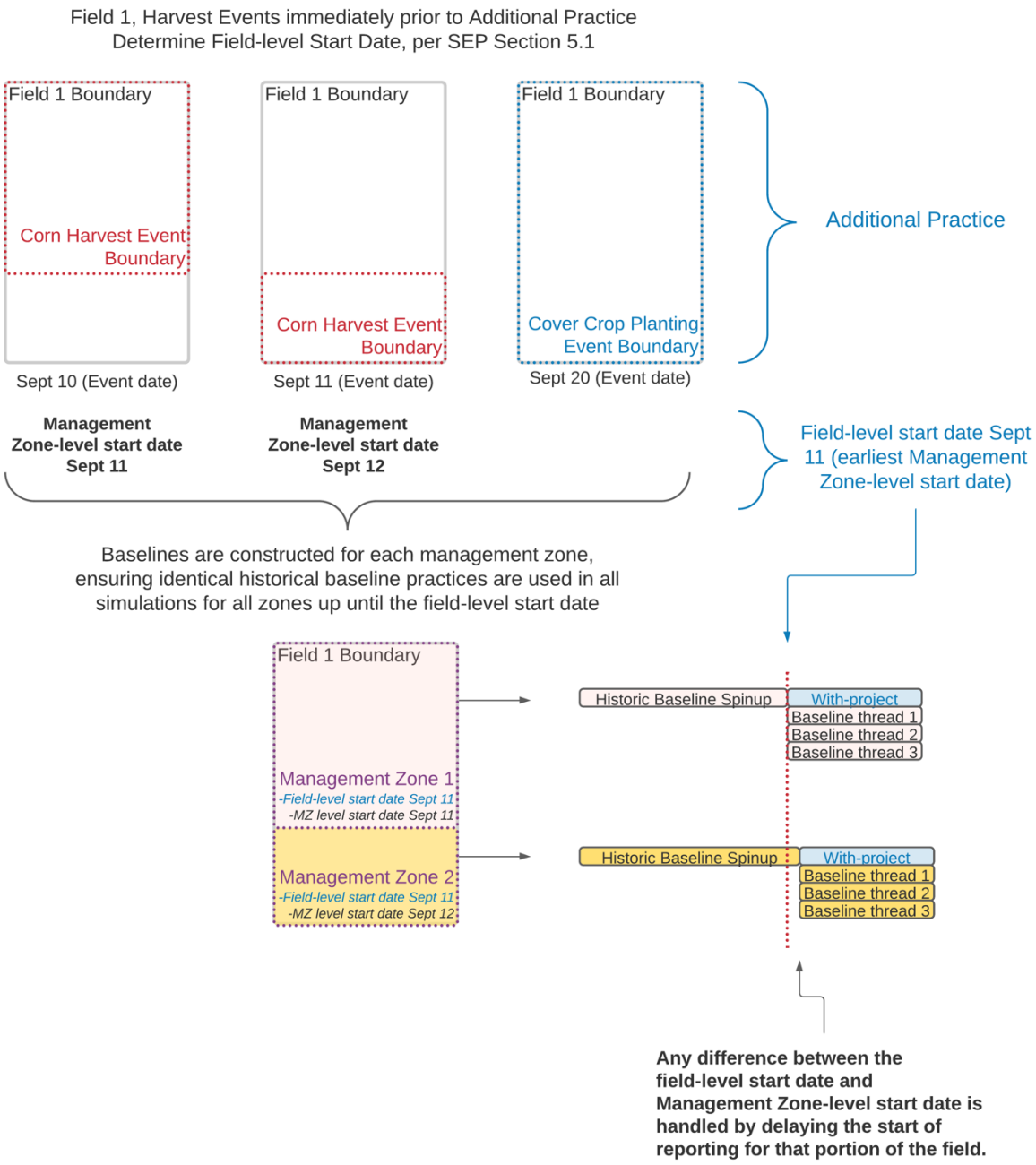


Figure 5: Example of how temporally separate harvest events immediately prior to the adoption of an additional practice impact the field-level start date, versus the Management Zone-level start date, and how these dates are used to construct baselines.

some of these issues. Additional software improvements are being developed to fully resolve all discrepancies for the next reporting period.

6 Constructing modeled baselines for individual sample points (Step 5, Fig 1)

Baseline simulations and with-project simulations are constructed at the level of Management Zones for all sample points with that zone (Figure 4). For an individual sample point, baseline simulations extend from the exact same spinup simulations, which use all required historical baseline period data and additional sources (see IndigoCarbon_US-1_2023_0073) to meet DayCent-CR Version 1.1.0 input requirements as described in the Validation Report (see IndigoCarbon_US-1_2023_0046). This means: each baseline simulation as well as the with-project simulation have the same model-specific spinup and required historical baseline period spinup, such that all simulations have the same value up until the ***field-level start date.*** See Figure 9 for a diagram of how baselines are constructed following spinup requirements for DayCent-CR Version 1.1.0.

In fields with >1 Management Zone, simulations may continue to have the same value after the ***field-level start date***, up until the ***Management Zone-level start date***, in the event there is a discrepancy (see Figure 5 for example)**.**

Historical cultivation cycles are used to create unique historical segments (example provided in Figure 7) to start parallel baseline threads. Within each baseline thread, per SEP Section 5.1 requirements, management events from the required historical baseline period are rotated continuously from different starting points such that all historical management events are modeled for each reporting period (see Figure 9 for example). Historical segments are defined using crop growing seasons in order to retain agronomic realism to the greatest extent possible (see Figure 7).

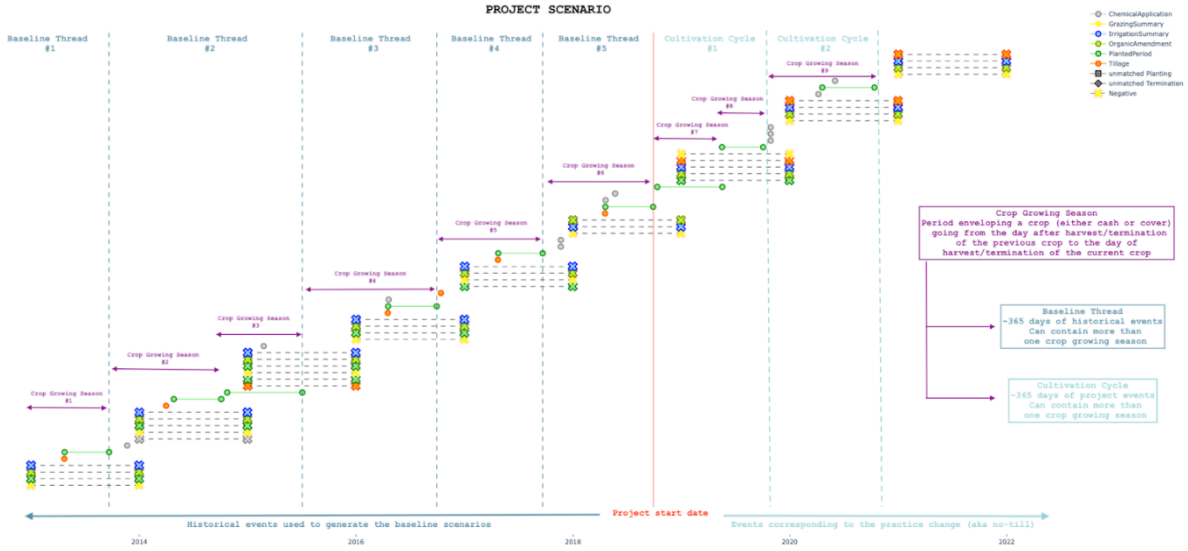


Figure 6: Example of breaking historical events into baseline threads (blue text at top label thread breaks, which occur at dotted lines to the left of the red line) based on crop growing seasons (marked with purple arrows) within historical cultivation cycles. Two with-project cultivation cycles also shown, starting at red lines and going to right.

Note, in the following section describing the steps of baseline construction logic, ***the term baseline threads is used as it is applied in code,*** wherein baseline threads are the list of historical segments, as defined above, and manipulated as needed to meet SEP Section 5.1 requirements. See Figure 9 for a diagram of how baseline construction is done for DayCent-CR Version 1.1.0.

Baseline logic:

1. Baselines are constructed for each management zone.
2. For a given simulated soil sample point (for initial measured SOC), the with-project simulation as well as each simulated baseline will use the same information to spin-up and initialize the model, such that the value of all

simulations is the same up to the field-level start date.

3. Baseline threads segment all historical data provided by the grower into non-overlapping periods of time using crop growing seasons to determine historical cultivation cycles (see Figure 7 for example).
4. The number of baseline threads will be, at a minimum, the number of years (in 365-day periods) of the required historical baseline period and may exceed this number if a grower has provided sufficient additional data.
5. Each unique baseline thread will be used to start a unique baseline simulation.
6. Baseline threads will be rotated in order continuously from whichever thread is used to start an individual baseline simulation.
7. Each baseline thread will start simulating historical crop practices on the first day of the first crop growing season or fallow season that begins in that historical cultivation cycle.
8. For the first historical cultivation cycle (i.e. the earliest), the crop growing season starts the day before the first event, whether that event is positive (i.e. grower attested that a practice occurred) or negative (i.e grower attested that no practice occurred).
9. If possible, each individual baseline simulation begins on the ***additionality_start_date*** (code reference for the field-level start date). If there is a discrepancy between the ***additionality_start_date*** and the ***period_start*** (code reference for the Management Zone-level start date), individual baseline simulations will begin the day of the ***period_start*** (see Figure 5 for example).
10. To start an individual baseline simulation, the following logic is used:
    - First, attempt to maintain the calendar date of each event in the baseline thread starting that baseline simulation. The counterfactual historical cultivation cycle is placed in the with-project period such that there is no overlap of previous planted periods for the field. Calendar dates (i.e. day of the year) are preserved for each event within the historical cultivation cycle.
      - Example: using a historical corn planting event from May 1, 2018 to generate a similar event in the project period of the baseline scenario in 2022, by moving the planting event to *May 1* of 2022.
    - If it is not possible to maintain the calendar date of each event because the initial baseline thread events conflict with the concluding historical cultivation cycle used for model-spinup, conflicts are resolved as follows:
      - Concluding historical cultivation cycle crop practices are preserved unchanged, as these events best reflect agronomic reality.
      - If the conflict of event dates is regarding the first planting date of the baseline thread:, the planting date of the historical cycle is moved to the day after the harvest of the concluding growing season (see Figure 8, A);
      - If the conflict of event dates is regarding field preparation *before* the first planting date of the baseline thread:
        * First, an attempt is made to move the conflicting events to start the day after the ***period_start*** (see Figure 8, B1). This move is made if there is sufficient time without events between the ***period_start*** and the first planting date of the baseline thread. In this case, conflicting events are moved to that time period.
        * If there is no time between the ***period_start*** and the first planting date of the baseline thread, or if there is not enough time that all conflicting events can fit while retaining the relative amount of time between the events, conflicting events get clipped (see Figure 8, B2). The approach ensures that as many counterfactual events are modeled as possible, while still preserving agronomic realism by retaining the completion of crop growing seasons as well as the timing between events.
    - If the starting baseline thread is seasonally separated, a fallow season is generated (see Figure 8, C).
11. All constructed baselines will be submitted to be modeled for each monitoring report, regardless of whether the crop rotation being followed on a field allows for the use of the matched versus blended baselines. We only blend baseline simulation results for every modeled location, and in all circumstances calculate the average

using all threads rather than using the subset of matched baselines. See also IndigoCarbon_US-1_2023_0065 for additional details on this step in model post-processing.

Regarding item 11: we chose to never use matched baselines after we identified several potential issues in the interpretation of matched baseline requirements in the SEP. In exploring these issues, we determined that using matched baselines vastly increases complexity of requirements without having a net impact on quantification of emission reductions. Further, these complexities greatly increased with the use of management zones, which otherwise provide the most accurate approach to model baseline practices as they were spatially implemented within a field. See Section 5 below for full details of our analysis.

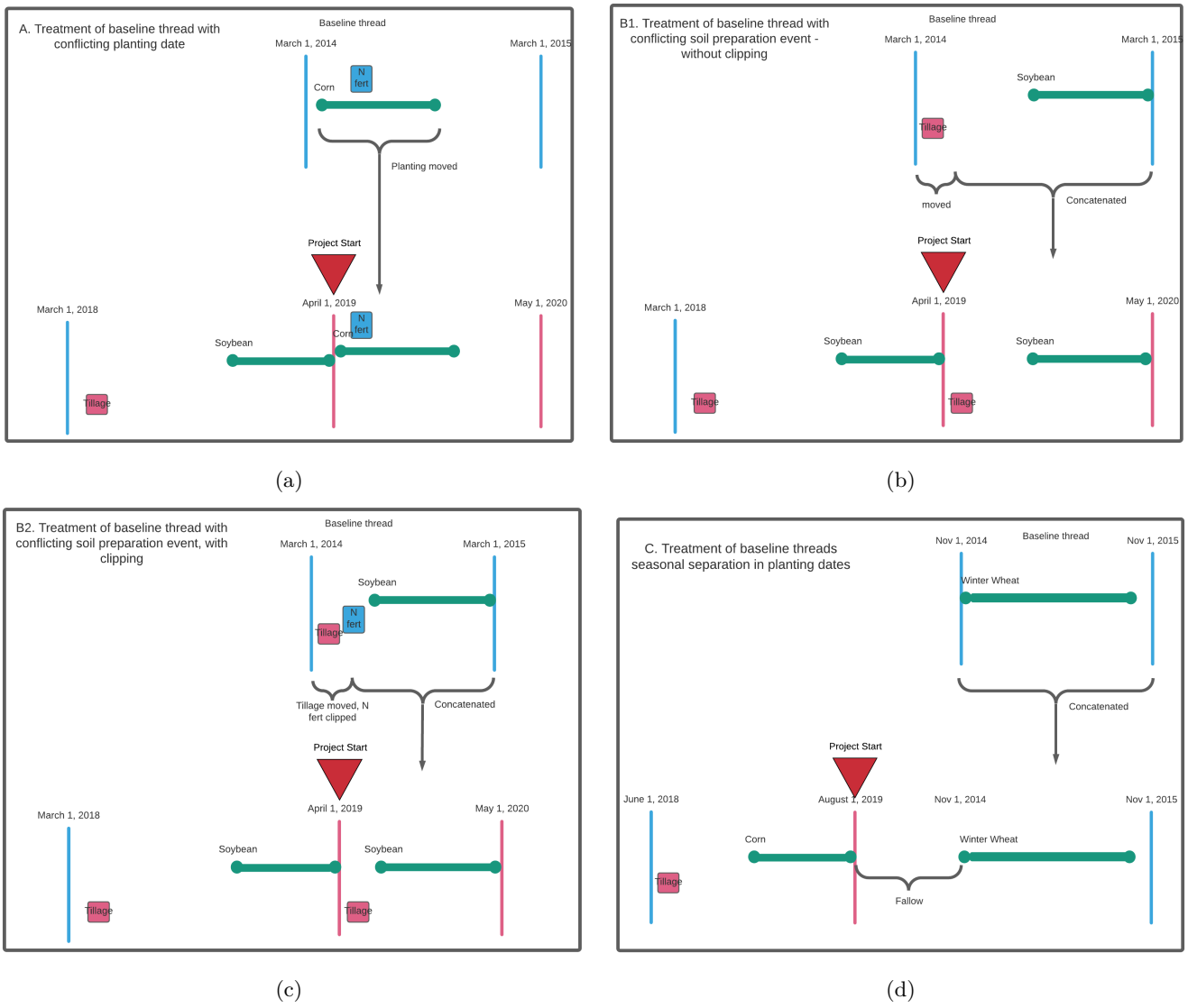


Figure 7: Treatment of historical threads under scenarios of conflicting dates of events (A and B), and scenarios where the planting date is seasonally separated (C)

7 Reasoning for only using blended baselines

First, we identified two potential issues in interpreting the SEP for matched baseline requirements:

Issue 1: Conflicting SEP requirements in Section 3.4.1.4 for matched baselines

SEP text reads *"If the current year's crop matches with the same crop in more than one of the parallel baseline threads, then the matched baseline is an average of the results for those threads for that year."* However, the left bottom white

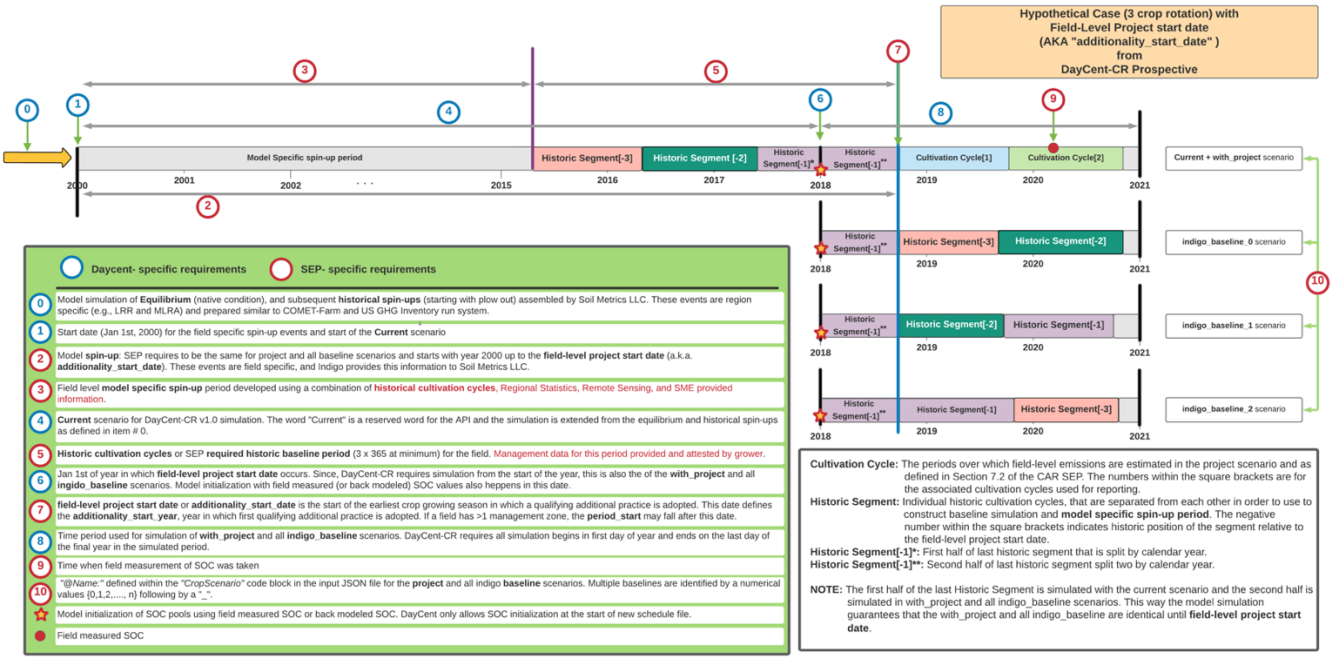


Figure 8: Baseline construction, following SEP requirements, conforming to DayCent-CR Version 1.1.0 model structure

square of Figure 3.2, Box 3, which has an additional requirement of having at least 2 years of historical baseline data about that individual crop. We evaluated the interpretation following the text rather than the figure. The result of following the figure would be further limiting the circumstances in which matched baselines are an option.

Issue 2: SEP Interpretation considering multiple cultivation cycles

SEP protocol does not consider reporting periods that extend across multiple cultivation cycles with multiple cash crops. Section 3.4.1.4 is written based on each reporting period having only one crop type. SEP requirements do not make it clear whether the unit to determine matching is *all cash crops* within a reporting period, or whether the unit is *each individual cash crop* within a reporting period. An alternative would be to use matched baselines per cultivation cycle within the reporting period. To show the difference:

Table 1: Hypothetical impacts on the use of matching for a single field rotating soy (S), wheat (W), and corn (C), given two different SEP interpretations.

Historical baseline rotation	**With-project cultivation cycles**	**Interpretation 1: Reporting Period is the unit to define matching**	**Interpretation 2: Cultivation cycle is the unit to define matching**
For S- W - C - S - C	S - W	Average across one baseline S- W - C - S - C- **S - W** with project S- W - C - S - C - **S - W** baseline 0 match S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - S - C baseline 3 S- W - C - S - C- C - S baseline 4	1st cultivation cycle, average across S S- W - C - S - C - **S** - W with project S- W - C - S - C - **S** - W baseline 0 match S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - **S** - C baseline 3 match S- W - C - S - C - C - S baseline 4 2nd cultivation cycle, average across only W S- W - C - S - C - S - **W** with project S- W - C - S - C - S - **W** baseline 0 match S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - S - C baseline 3 S- W - C - S - C - C - S baseline 4
For S- W - C - S - C	S - C	No matches, use blended baseline S- W - C - S - C- **S - C** with project S- W - C - S - C -S - W baseline 0 S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - S - C baseline 3 S- W - C - S - C- C - S baseline 4	1st cultivation cycle, average across S S- W - C - S - C - **S** - C with project S- W - C - S - C - **S** - W baseline 0 match S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - **S** - C baseline 3 match S- W - C - S - C - C - S baseline 4 2nd cultivation cycle, switch to blended S- W - C - S - C - S - C with project S- W - C - S - C - S - W baseline 0 S- W - C - S - C - W - C baseline 1 S- W - C - S - C - C - S baseline 2 S- W - C - S - C - S - C baseline 3 S- W - C - S - C - C - S baseline 4

We chose to construct an analysis to explore the 2nd interpretation, as the impact of the 1st interpretation would only be to further limit the frequency of matched baselines.

Analysis of matched baseline impacts on emission source quantification

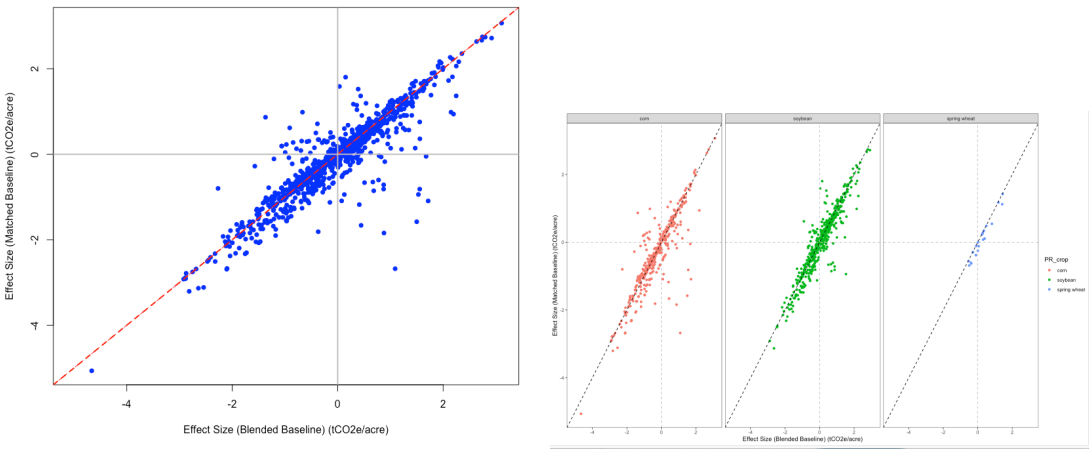
We used a test run of ~600 fields modeled on August 16, 2021 using the validated version of DayCent-CR Version 1.0 but a version of the API that was still in the process of debugging, which means that the results are not exactly the same as in production runs for this report. However, API debugging since these simulations have not greatly impacted overall simulation results. We used modeled results from the test run and quantified credits using a version of Gate 7 Level 1 code (model post-processing) that had not yet been productionized and debugged. However, we would not expect the final productionized script to fundamentally change the pattern of results observed in in this analysis, as the version of the post-processing script used to generate these results is applied to both project and baseline scenarios, and so would not bias one over the other.

From the test run, a subset of 114 fields and 949 points generated emission source change estimates for SOC (that met criteria for using a matched approach) were quantified using Interpretation 2 (Table 1). For the majority of fields, the impact on ER was minimal (clustering at 1:1 lines in Figure 9). There is some scatter, indicating that for some points a matched approach does change ER. However, the scatter is relatively symmetric on either side of the 1:1 line, indicating that a matched approach can go in either direction in terms of impacting ER. While only a subset of fields

were included in this analysis, their inclusion was random. We would expect similar results with a more complete representation of our grower population.

Conclusions:

- Effect sizes, at the project level, don't greatly differ whether matched AND blended baselines or ONLY blended are applied.
- Matched baseline requirements are less straightforward to implement than blended baselines that make equal use of all data from the historical baseline period.
- Due to their complexity, and potentially due to other interpretations of SEP requirements than the one evaluated here, relatively few fields will qualify to use the matched approach (and those that do are unlikely to qualify beyond the first few cultivation cycles).
- Therefore, we only implemented blended baselines for all modeled points in the project.



(a) (b)

Figure 9: Comparison of Effect Size using blended baselines (x axis) versus matched baselines (y axis) 949 points in 114 fields overall (top panel), and by crop type (corn in red, soybean in green, and spring wheat in blue,)(b), showing clustering around 1:1 line.

8 Summary of Changes

Section	Date	Summary of Changes
-	26 June 2024	Meeting field-level SEP requirements for baseline construction
	26 June 2024	Reference to monthly timestep updated to reflect new modeling approach